



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	Bunge T-5	Unit ID	T-5
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured	11-N	Primary KV	12.470
Gallons	333	KVA Rating	1500
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/06/2026	08/30/2019
Barcode Sample Number	00394455	00394433	00091838
Identification	T-5 Bunge	T-5 Bunge	
Identification	T-5 Bunge	T-5 Bunge	
DGA ppm			
Hydrogen	27	17	12
Methane	5	22	11
Ethane	302	4	187
Ethylene	6	3	3
Acetylene	0	0	0
Carbon Monoxide	71	1272	77
Carbon Dioxide	889	7141	499
Oxygen	904	4376	282
Nitrogen	94205	100876	70558
TDCG	411	1318	290
TDCG Rate ppm / day	-907.0	0.4	-0.1
CO2 / CO	12.52	5.61	6.48
DGA Condition	Condition 2	Condition 3	Condition 1
DGA Analysis	Per C57.155-2014, Ethane gas is at a condition 2 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer may have Thermal Faults (< 300 C).	Per C57.155-2014, Methane gas is at a condition 2 level and recommend investigating the source of activity. Per C57.155-2014, Carbon Monoxide gas is at a condition 3 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).	
Operating Procedures	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is in a condition 3 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.
Sampling Interval	Annually	Monthly	Annually
Barcode Sample Number	00394456	00394434	00091839
Moisture (D1533) ppm	2	2	15
Acid (D974) mgKOH/g	0.047	0.002	
IFT mM/m	18.1	39.7	
Color (D1500)	0.25	0.75	
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - Trace Sediment	
Dielectric (D1816) kV 2 mm	46.4	54.2	65.1
Specific Gravity (D1298/D4052)	0.9221	0.8889	
PF 25C (D924) %	0.209	0.020	
PCB (D4059) ppm	< 1	< 1	
Oil Classification		Class I	

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	Bunge T-5	Unit ID	T-5
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured	11-N	Primary KV	12.470
Gallons	333	KVA Rating	1500
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/06/2026	08/30/2019
Oil Quality	All tests meet recommended levels for in-service natural ester fluids.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	970251-A3	Unit ID	XF-3
Equipment Type	Transformer	Manufacturer	Square D
Year Manufactured	7/97	Primary KV	12.470
Gallons	445	KVA Rating	2000/2240
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/07/2025	04/20/2023
Barcode Sample Number	00394439	00336416	00217049
Job Number		25079	
Identification	XF-3 Bunnge		
Identification	XF-3 Bunge		
DGA ppm			
Hydrogen	9	0	9
Methane	31	28	28
Ethane	7	5	5
Ethylene	28	24	26
Acetylene	0	0	0
Carbon Monoxide	947	851	843
Carbon Dioxide	5511	4995	5179
Oxygen	5473	1679	1585
Nitrogen	116307	77580	91786
TDCG	1022	908	911
CO2 / CO	5.82	5.87	6.14
DGA Condition	Status 2	Status 1	Status 1
DGA Analysis	Per C57.104-2019, Carbon Monoxide gas is in a status 2 condition and recommend investigating the source of activity. Per C57.104-2019, the transformer is a status 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present. Duval Pentagon indicates the transformer may have Thermal Faults (300 - 700 C).	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable.	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable.
Operating Procedures	Per C57.104-2019, the combustible gas level warrants additional oil testing and analysis. Exercise caution and evaluate transformer load conditions.	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.
Sampling Interval	Annually	Annually	Annually
Barcode Sample Number	00394440	00336417	00217050
Moisture (D1533) ppm	8	7	3
Acid (D974) mgKOH/g	0.005	0.005	0.006
IFT mM/m	41.6	39.9	40.2
Color (D1500)	0.75	1.00	1.00
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment	Clear and Sparkle - Trace Sediment
Dielectric (D1816) kV 2 mm	50.0	60.2	54.2
Specific Gravity (D1298/D4052)	0.8892	0.8893	0.8900
PF 25C (D924) %	0.059	0.023	0.004
PCB (D4059) ppm	< 1		
Oil Classification	Class I	Class I	Class I

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	970251-A3	Unit ID	XF-3
Equipment Type	Transformer	Manufacturer	Square D
Year Manufactured	7/97	Primary KV	12.470
Gallons	445	KVA Rating	2000/2240
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/07/2025	04/20/2023
Oil Quality	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	BUNGE T-7	Unit ID	T-7
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured	11-N	Primary KV	12.47
Gallons	333	KVA Rating	1500
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/11/2025	05/25/2021
Barcode Sample Number	00394445	00336420	00147446
Job Number		25079	
Identification	T-7 Bunge		
Identification	T-7 Bunge		
DGA ppm			
Hydrogen	11	0	15
Methane	47	34	34
Ethane	36	28	28
Ethylene	6	5	7
Acetylene	0	0	0
Carbon Monoxide	687	643	851
Carbon Dioxide	3739	3507	3703
Oxygen	381	637	521
Nitrogen	81693	70779	95335
TDCG	787	710	935
CO2 / CO	5.44	5.45	4.35
DGA Condition	Status 1	Status 1	Status 3
DGA Analysis	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable.	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable.	Per C57.104-2019, Carbon Monoxide gas is in a status 1 condition but the rate of change has increased. Recommend investigating the source of activity. Per C57.104-2019, the rate of change of gas levels indicates the transformer is in a status 3 state with an accelerated rate of decomposition. Recommend re-testing to establish a trend as faults are probably present. Duval Pentagon indicates the transformer may have Thermal Faults (< 300 C).
Operating Procedures	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.	Per C57.104-2019, the rate of increase of combustible gases warrant additional oil testing and analysis. Exercise caution and potentially plan for an outage for inspection, electrical testing and consultation.
Sampling Interval	Annually	Annually	Monthly
Barcode Sample Number	00394446	00336421	00147447
Moisture (D1533) ppm	12	7	7
Acid (D974) mgKOH/g	0.024	0.016	0.025
IFT mM/m	30.0	29.9	28.8
Color (D1500)	1.00	1.00	1.75
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - Trace Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	55.0	43.5	53.7

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	BUNGE T-7	Unit ID	T-7
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured	11-N	Primary KV	12.47
Gallons	333	KVA Rating	1500
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/11/2025	05/25/2021
Specific Gravity (D1298/D4052)	0.8933	0.8933	0.8940
PF 25C (D924) %	0.068	0.065	0.074
PCB (D4059) ppm	< 1		
Oil Classification	Class I	Class I	Class I
Oil Quality	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	195013621	Unit ID	T-1
Equipment Type	Transformer	Manufacturer	Eaton
Year Manufactured		Primary KV	12.47
Gallons	469	KVA Rating	2500
Phases	3	Fluid type	Natural Ester
Substation Location	Bunge	Breathing Configuration	Free Breather

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/08/2025	04/17/2023
Barcode Sample Number	00394447	00336428	00217045
Job Number		25079	23108
Identification	T-1 Bunge		
Identification	T-1 Bunge		
DGA ppm			
Hydrogen	99	77	14
Methane	8	7	9
Ethane	719	459	845
Ethylene	13	9	18
Acetylene	2	2	5
Carbon Monoxide	66	44	69
Carbon Dioxide	2329	1541	1414
Oxygen	3048	873	1131
Nitrogen	84972	44009	61338
TDCG	907	598	960
TDCG Rate ppm / day	0.9	-0.4	0.5
CO2 / CO	35.29	35.02	20.49
DGA Condition	Condition 3	Condition 3	Condition 4
DGA Analysis	Per C57.155-2014, Acetylene gas is at a condition 2 level and recommend investigating the source of activity. Per C57.155-2014, Ethane gas is at a condition 3 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).	Per C57.155-2014, Acetylene gas is at a condition 2 level and recommend investigating the source of activity. Per C57.155-2014, Ethane gas is at a condition 3 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).	Per C57.155-2014, Ethylene gas is at a condition 2 level and recommend investigating the source of activity. Per C57.155-2014, Acetylene gas is at a condition 3 level and recommend investigating the source of activity. Per C57.155-2014, Ethane gas is at a condition 4 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).
Operating Procedures	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is in a condition 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is in a condition 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is in a condition 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present.
Sampling Interval	Quarterly	Annually	Quarterly
Barcode Sample Number	00394448	00336429	00217046
Moisture (D1533) ppm	44	57	41
Acid (D974) mgKOH/g	0.048	0.049	0.034
IFT mM/m	22.0	21.3	21.1
Color (D1500)	0.50	1.00	1.00
Visual (1524)	Clear and Sparkle - Trace Sediment	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	58.6	68.1	66.0
Specific Gravity (D1298/D4052)	0.9225	0.9230	0.9246

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	195013621	Unit ID	T-1
Equipment Type	Transformer	Manufacturer	Eaton
Year Manufactured		Primary KV	12.47
Gallons	469	KVA Rating	2500
Phases	3	Fluid type	Natural Ester
Substation Location	Bunge	Breathing Configuration	Free Breather

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date		07/06/2026	07/08/2025	04/17/2023
PF 25C (D924)	%	0.619	0.541	0.598
PCB (D4059)	ppm	< 1		
Oil Quality		All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	L463887T74AA	Unit ID	PUMP HOUSE
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured		Primary KV	12.470
Gallons	125	KVA Rating	225
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/07/2025	04/18/2023
Barcode Sample Number	00394449	00336440	00208107
Job Number		25079	
Identification	T-FP Bunge		
Identification	T-FP Bunge		
DGA ppm			
Hydrogen	11	41	28
Methane	59	57	59
Ethane	178	174	162
Ethylene	5	3	6
Acetylene	0	0	0
Carbon Monoxide	396	775	752
Carbon Dioxide	3661	4271	4671
Oxygen	622	615	564
Nitrogen	68982	80322	83022
TDCG	649	1050	1007
CO2 / CO	9.24	5.51	6.21
DGA Condition	Status 3	Status 2	Status 2
DGA Analysis	Per C57.104-2019, Ethane gas is in a status 3 condition and recommend investigating the source of activity. Per C57.104-2019, the transformer is in a status 3 state with a high rate of decomposition. Recommend re-testing to establish a trend as faults are probably present. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).	Per C57.104-2019, Ethane gas is in a status 2 condition and recommend investigating the source of activity. Per C57.104-2019, the transformer is in a status 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present. Duval Pentagon indicates the transformer may have Stray Gassing of Oil (<200 C).	Per C57.104-2019, Ethane gas is in a status 2 condition and has stabilized. Recommend investigating the source of activity. Per C57.104-2019, the change of gas levels from the last sample indicates the transformer is in a status 2 state with greater than normal combustible gas activity. Recommend establishing a trend as faults may be present. Duval Pentagon indicates the transformer may have Stray Gassing of Oil (<200 C).
Operating Procedures	Per C57.104-2019, the combustible gas levels warrant additional oil testing and analysis. Exercise caution and potentially plan for an outage for inspection, electrical testing and consultation.	Per C57.104-2019, the combustible gas level warrants additional oil testing and analysis. Exercise caution and evaluate transformer load conditions.	Per C57.104-2019, the combustible gas level warrants additional oil testing and analysis. Exercise caution and evaluate transformer load conditions.
Sampling Interval	Quarterly	Annually	Quarterly
Barcode Sample Number	00394450	00336441	00208108
Moisture (D1533) ppm	2	9	3
Acid (D974) mgKOH/g	0.025	0.017	0.026
IFT mM/m	30.1	29.2	29.8
Color (D1500)	1.00	1.75	1.50
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	50.9	38.7	70.9
Specific Gravity (D1298/D4052)	0.8987	0.8987	0.8983

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	L463887T74AA	Unit ID	PUMP HOUSE
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured		Primary KV	12.470
Gallons	125	KVA Rating	225
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/07/2025	04/18/2023
PF 25C (D924) %	0.158	0.166	0.180
PCB (D4059) ppm	< 1		
Oil Classification	Class I	Class II	Class I
Oil Quality	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	Dielectric 1816 2 mm test did not meet the C57.106-2015 minimum standard. Oil Is considered a Class II oil and recommend reconditioning the oil.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	195013620	Unit ID	T-2
Equipment Type	Transformer	Manufacturer	Eaton
Year Manufactured		Primary KV	12.47
Gallons	469	KVA Rating	2500
Phases	3	Fluid type	Natural Ester
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/11/2025	04/17/2023
Barcode Sample Number	00394453	00336422	00217047
Job Number		25079	23108
Identification	T-2 Bunge		
Identification	T-2 Bunge		
DGA ppm			
Hydrogen	55	36	25
Methane	10	9	5
Ethane	722	444	429
Ethylene	20	12	8
Acetylene	5	5	1
Carbon Monoxide	86	54	57
Carbon Dioxide	5405	3370	1392
Oxygen	759	544	558
Nitrogen	83701	43538	44281
TDCG	898	560	525
TDCG Rate ppm / day	0.9	0.0	-1.3
CO2 / CO	62.85	62.41	24.42
DGA Condition	Condition 3	Condition 3	Condition 3
DGA Analysis	Per C57.155-2014, Ethylene gas is at a condition 2 level and recommend investigating the source of activity. Per C57.155-2014, Ethane and Acetylene gases are at a condition 3 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).	Per C57.155-2014, Ethane and Acetylene gases are at a condition 3 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).	Per C57.155-2014, Acetylene gas is at a condition 2 level and recommend investigating the source of activity. Per C57.155-2014, Ethane gas is at a condition 3 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).
Operating Procedures	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is in a condition 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is in a condition 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.
Sampling Interval	Quarterly	Quarterly	Annually
Barcode Sample Number	00394454	00336423	00217048
Moisture (D1533) ppm	52	49	44
Acid (D974) mgKOH/g	0.066	0.055	0.029
IFT mM/m	16.9	16.1	17.7
Color (D1500)	0.75	0.50	0.50
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	67.3	72.4	69.9
Specific Gravity (D1298/D4052)	0.9223	0.9231	0.9244
PF 25C (D924) %	0.562	0.453	0.749
PCB (D4059) ppm	< 1		

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	195013620	Unit ID	T-2
Equipment Type	Transformer	Manufacturer	Eaton
Year Manufactured		Primary KV	12.47
Gallons	469	KVA Rating	2500
Phases	3	Fluid type	Natural Ester
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/11/2025	04/17/2023
Oil Quality	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	M11J12001	Unit ID	XF-6
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured	11-1	Primary KV	12.470
Gallons	333	KVA Rating	1500/1680
Phases	3	Fluid type	Natural Ester
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/07/2025	04/20/2023
Barcode Sample Number	00394461	00336446	00217041
Job Number		25079	
Identification	XF-6 Bunge		
Identification	XF-6 Bunge		
DGA ppm			
Hydrogen	14	13	13
Methane	8	13	11
Ethane	196	179	182
Ethylene	3	4	3
Acetylene	0	0	0
Carbon Monoxide	56	68	69
Carbon Dioxide	927	1055	1111
Oxygen	779	1026	416
Nitrogen	61734	76133	78300
TDCG	277	277	278
TDCG Rate ppm / day	0.0	0.0	0.1
CO2 / CO	16.55	15.51	16.10
DGA Condition	Condition 1	Condition 1	Condition 1
DGA Analysis	Duval Pentagon not applicable.	Duval Pentagon not applicable.	Duval Pentagon not applicable.
Operating Procedures	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.
Sampling Interval	Annually	Annually	Annually
Barcode Sample Number	00394462	00336447	00217042
Moisture (D1533) ppm	56	73	35
Acid (D974) mgKOH/g	0.040	0.043	0.040
IFT mM/m	22.2	21.6	22.2
Color (D1500)	0.25	1.00	0.50
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	66.5	71.1	57.6
Specific Gravity (D1298/D4052)	0.9205	0.9222	0.9229
PF 25C (D924) %	0.255	0.231	0.189
PCB (D4059) ppm	< 1		
Oil Quality	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	1950013619	Unit ID	T-9
Equipment Type	Transformer	Manufacturer	Eaton
Year Manufactured		Primary KV	12.47
Gallons	469	KVA Rating	2500
Phases	3	Fluid type	Natural Ester
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/11/2025	04/17/2023
Barcode Sample Number	00394469	00336426	00165376
Job Number		25079	
Identification	T-9 Bunge		
Identification	T-9 Bunge		
DGA ppm			
Hydrogen	27	41	23
Methane	7	11	6
Ethane	439	395	356
Ethylene	14	14	10
Acetylene	6	7	2
Carbon Monoxide	67	84	75
Carbon Dioxide	1691	1671	963
Oxygen	1228	960	839
Nitrogen	63578	67036	54393
TDCG	560	552	472
TDCG Rate ppm / day	0.0	0.1	-1.1
CO2 / CO	25.24	19.89	12.84
DGA Condition	Condition 3	Condition 3	Condition 2
DGA Analysis	Per C57.155-2014, Ethane and Acetylene gases are at a condition 3 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).	Per C57.155-2014, Ethane and Acetylene gases are at a condition 3 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer has Thermal Faults (< 300 C).	Per C57.155-2014, Ethane and Acetylene gases are at a condition 2 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer may have Thermal Faults (< 300 C).
Operating Procedures	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is in a condition 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is in a condition 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.
Sampling Interval	Quarterly	Quarterly	Annually
Barcode Sample Number	00394470	00336427	00165377
Moisture (D1533) ppm	39	48	49
Acid (D974) mgKOH/g	0.037	0.039	0.027
IFT mM/m	14.8	14.6	15.7
Color (D1500)	0.25	0.50	0.50
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	53.6	65.4	67.1
Specific Gravity (D1298/D4052)	0.9222	0.9231	0.9242
PF 25C (D924) %	0.312	0.319	0.420
PCB (D4059) ppm	< 1		
Oil Quality	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	M11J12002	Unit ID	XF-5
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured	11-1	Primary KV	12.470
Gallons	333	KVA Rating	1500/1680
Phases	3	Fluid type	Natural Ester
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/07/2025	04/20/2023
Barcode Sample Number	00394473	00336444	00217039
Job Number		25079	
Identification	XF-5 Bunge		
DGA ppm			
Hydrogen	0	8	0
Methane	11	10	14
Ethane	220	206	189
Ethylene	3	4	3
Acetylene	0	0	0
Carbon Monoxide	49	62	55
Carbon Dioxide	882	987	908
Oxygen	4098	805	721
Nitrogen	77309	77555	66944
TDCG	283	290	261
TDCG Rate ppm / day	0.0	0.0	0.0
CO2 / CO	18.00	15.92	16.51
DGA Condition	Condition 1	Condition 1	Condition 1
DGA Analysis	Duval Pentagon not applicable.	Duval Pentagon not applicable.	Duval Pentagon not applicable.
Operating Procedures	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.
Sampling Interval	Annually	Annually	Annually
Barcode Sample Number	00394474	00336445	00217040
Moisture (D1533) ppm	32	81	22
Acid (D974) mgKOH/g	0.038	0.043	0.038
IFT mM/m	22.1	21.6	21.9
Color (D1500)	0.25	1.00	0.50
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - Trace Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	63.9	67.6	65.8
Specific Gravity (D1298/D4052)	0.9209	0.9222	0.9212
PF 25C (D924) %	0.358	0.343	0.259
PCB (D4059) ppm	< 1		
Oil Quality	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.	All tests meet recommended levels for in-service natural ester fluids.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	ES121318PADBUN3	Unit ID	
Equipment Type	Transformer	Manufacturer	
Year Manufactured		Primary KV	.480
Gallons	293	KVA Rating	750
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	Free Breather

Equipment Information		
Top Valve (in)		Bottom Valve (in)
Hose Length (ft)		Paint Condition
Conservator Tank		Bushings Enclosed
Leaks		Radiators
Service Energized		Compartments

Sample Date	07/06/2026	08/08/2020	06/26/2019
Barcode Sample Number	00394489	00124283	00059755
Job Number	26073		
Identification	T 2000 Bunge		
Identification	T 2000 Bunge		
DGA ppm			
Hydrogen	55		0
Methane	266		30
Ethane	104		10
Ethylene	536		0
Acetylene	3		0
Carbon Monoxide	594		5
Carbon Dioxide	2459		425
Oxygen	18695		33562
Nitrogen	82571		69801
TDCG	1558		45
CO2 / CO	4.14		85.00
DGA Condition	Status 3		Status 2
DGA Analysis	Per C57.104-2019, Carbon Dioxide gas is at a status 1 condition but has increased. Recommend investigating the source of activity. Per C57.104-2019, Acetylene, Carbon Monoxide and Hydrogen gases are in a status 2 condition and have increased. Recommend investigating the source of activity. Per C57.104-2019, Ethylene, Ethane and Methane gases are in a status 3 condition and have increased. Recommend investigating the source of activity. Per C57.104-2019, the change of gas levels from the previous sample indicates the transformer is in a status 3 state with a high rate of decomposition. Recommend immediate re-testing to establish a trend as faults are probably present. Duval Pentagon indicates the transformer has Thermal Faults (> 300 C).		Per C57.104-2019, Methane gas is in a status 2 condition and recommend investigating the source of activity. Per C57.104-2019, the transformer is a status 2 state with greater than normal combustible gas level. Recommend establishing a trend as faults may be present. Duval Pentagon indicates the transformer may have Thermal Faults (< 300 C).
Operating Procedures	Per C57.104-2019, the increase combustible gases warrant additional oil testing and analysis. Exercise extreme caution, plan for an outage for inspection, electrical testing and consultation.		Per C57.104-2019, the combustible gas level warrants additional oil testing and analysis. Exercise caution and evaluate transformer load conditions.
Sampling Interval	Monthly		Quarterly
Barcode Sample Number	00394490		00059756

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	ES121318PADBUN3	Unit ID	
Equipment Type	Transformer	Manufacturer	
Year Manufactured		Primary KV	.480
Gallons	293	KVA Rating	750
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	Free Breather

Equipment Information			
Top Valve (in)		Bottom Valve (in)	
Hose Length (ft)		Paint Condition	
Conservator Tank		Bushings Enclosed	
Leaks		Radiators	
Service Energized		Compartments	

Sample Date	07/06/2026	08/08/2020	06/26/2019
Moisture (D1533) ppm	6	30	31
Acid (D974) mgKOH/g	0.003	0.001	0.003
IFT mM/m	39.3	41.1	42.8
Color (D1500)	0.75	0.50	0.25
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle- No Sediment	Clear and Sparkle- No Sediment
Dielectric (D1816) kV 2 mm	49.8	43.3	49.3
Specific Gravity (D1298/D4052)	0.8848	0.8439	0.8417
PF 25C (D924) %	0.022	0.042	0.030
PCB (D4059) ppm	< 1		
Oil Classification	Class I	Class I	Class I
Oil Quality	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	L445634T73AA	Unit ID	T-4
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured	11-N	Primary KV	12.470
Gallons	510	KVA Rating	1000
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/07/2025	04/17/2023
Barcode Sample Number	00394491	00336414	00208109
Job Number		25079	
Identification	T-4 Bunge		
Identification	T-6 Bunge		
DGA ppm			
Hydrogen	18	27	18
Methane	38	33	38
Ethane	43	34	34
Ethylene	7	4	7
Acetylene	0	0	0
Carbon Monoxide	760	673	742
Carbon Dioxide	5629	4919	5973
Oxygen	1590	646	490
Nitrogen	101304	71750	83679
TDCG	866	771	839
CO2 / CO	7.41	7.31	8.05
DGA Condition	Status 1	Status 1	Status 1
DGA Analysis	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable. CO2/CO ratio indicates excessive overheating and or accelerated aging of cellulose insulation.	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable.	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable. CO2/CO ratio indicates excessive overheating and or accelerated aging of cellulose insulation.
Operating Procedures	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.
Sampling Interval	Annually	Annually	Annually
Barcode Sample Number	00394492	00336415	00208110
Moisture (D1533) ppm	16	6	2
Acid (D974) mgKOH/g	0.027	0.016	0.025
IFT mM/m	30.4	30.2	30.0
Color (D1500)	1.00	1.25	1.00
Visual (1524)	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	52.9	63.5	62.2
Specific Gravity (D1298/D4052)	0.8908	0.8907	0.8901
PF 25C (D924) %	0.077	0.082	0.074
PCB (D4059) ppm	< 1		
Oil Classification	Class I	Class I	Class I
Oil Quality	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	L453243T74AA	Unit ID	Bean Tank T-8
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured		Primary KV	12.470
Gallons	510	KVA Rating	1000
Phases	3	Fluid type	Mineral Oil
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	07/11/2025	04/17/2023
Barcode Sample Number	00394493	00336408	00208113
Job Number		25079	
Identification	T-8 Bunge		
Identification	T-8 Bunge		
DGA ppm			
Hydrogen	0	0	5
Methane	62	65	59
Ethane	40	34	34
Ethylene	20	19	21
Acetylene	0	0	0
Carbon Monoxide	463	453	494
Carbon Dioxide	4871	4933	5550
Oxygen	592	458	630
Nitrogen	80355	73563	80597
TDCG	585	571	613
CO2 / CO	10.52	10.89	11.23
DGA Condition	Status 1	Status 1	Status 1
DGA Analysis	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable.	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable.	Per C57.104-2019, the combustible gases indicates the transformer is operating properly. Duval Pentagon not applicable.
Operating Procedures	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.	Per C57.104-2019, the rate of combustible gas increase allows for continued normal operation.
Sampling Interval	Annually	Annually	Annually
Barcode Sample Number	00394494	00336409	00208114
Moisture (D1533) ppm	14	13	7
Acid (D974) mgKOH/g	0.037	0.029	0.041
IFT mM/m	28.0	27.7	27.6
Color (D1500)	1.50	2.00	1.75
Visual (1524)	Clear and Sparkle - Light Sediment	Clear and Sparkle - Trace Sediment	Clear and Sparkle - No Sediment
Dielectric (D1816) kV 2 mm	56.1	61.2	69.7
Specific Gravity (D1298/D4052)	0.8984	0.8982	0.8980
PF 25C (D924) %	0.107	0.120	0.119
PCB (D4059) ppm	< 1		
Oil Classification	Class I	Class I	Class I
Oil Quality	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.	All tests met C57.106-2015 minimum requirements and oil is considered a Class I for continued service.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com



AMP Quality Energy Services,
LLC
AMP Quality Energy Services,
LLC
4220 West Schrimsher SW Site
W1
Huntsville, AL, 35804

Nameplate Data			
Serial Number	Bunge T-6	Unit ID	T-6
Equipment Type	Transformer	Manufacturer	General Electric
Year Manufactured	11-N	Primary KV	12.470
Gallons	333	KVA Rating	1500
Phases	3	Fluid type	Natural Ester
Substation Location	Bunge	Breathing Configuration	

Equipment Information		
Top Valve (in)	Bottom Valve (in)	
Hose Length (ft)	Paint Condition	
Conservator Tank	Bushings Enclosed	
Leaks	Radiators	
Service Energized	Compartments	

Sample Date	07/06/2026	08/30/2019	05/11/2018
Barcode Sample Number	00394495	00091828	00052791
Identification	T-6 Bunge		
Identification	T-6 Bunge		
DGA ppm			
Hydrogen	106	13	18
Methane	32	7	11
Ethane	315	126	196
Ethylene	4	2	3
Acetylene	0	0	0
Carbon Monoxide	75	55	85
Carbon Dioxide	1433	469	815
Oxygen	441	1629	644
Nitrogen	82218	49105	69869
TDCG	532	203	313
TDCG Rate ppm / day	0.1	-0.2	
CO2 / CO	19.11	8.53	9.59
DGA Condition	Condition 2	Condition 1	Condition 1
DGA Analysis	Per C57.155-2014, Methane and Ethane gases are at a condition 2 level and recommend investigating the source of activity. Duval Pentagon indicates the transformer may have Thermal Faults (< 300 C).		
Operating Procedures	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.	The Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.	Per C57.155-2014, the Total Dissolved Combustible Gases (TDCG) indicates the transformer is operating properly.
Sampling Interval	Annually	Annually	
Barcode Sample Number	00394496	00091829	00052790
Moisture (D1533) ppm	44	24	35
Acid (D974) mgKOH/g	0.032		0.029
IFT mM/m	15.4		21.9
Color (D1500)	0.25		0.50
Visual (1524)	Clear and Sparkle - No Sediment		Clear and Sparkle- No Sediment
Dielectric (D1816) kV 2 mm	46.7	57.2	70.7
Specific Gravity (D1298/D4052)	0.9166		0.9188
PF 25C (D924) %	0.409		0.212
PCB (D4059) ppm	< 1		
Oil Quality	All tests meet recommended levels for in-service natural ester fluids.		All tests meet recommended levels for in-service natural ester fluids.

4588 Faircrest St SW, Canton, OH 44706

p: 330.498.6255 | f: 330.498.6261

www.mvadiagnostics.com